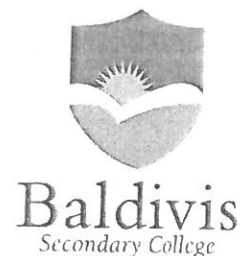


Name _____

Year 9

Class Solutions

Mini Test
Probability & Coordinate
Geometry



1. What is the theoretical probability of rolling a 2 on a die?

(1)

$\frac{1}{6}$



2. I tossed a coin 10 times to get the results;

H,T,H,H,T,H,T,H,H,H

Therefore, the $P(H) = 7/10$ is an example of **theoretical/experimental** probability.
Cross out the word that makes the statement incorrect.

(1)



3. Complete the following table for tossing two coins.

(3)

		Coin 2	
		H	T
Coin 1	H	H,H	H,T
	T	T,H	T,T



- (a) What is the probability of getting 2 tails?

$\frac{1}{4}$



4. Scrabble tiles have been placed in a hat. The only tiles in the hat make the word MATHEMATICS. A tile is selected randomly.
Determine the probability it will be;

(5)

- a) a M

$\frac{2}{11}$



- b) a vowel

$\frac{4}{11}$



- c) not a vowel

$\frac{7}{11}$



- d) an M or a C

$\frac{3}{11}$

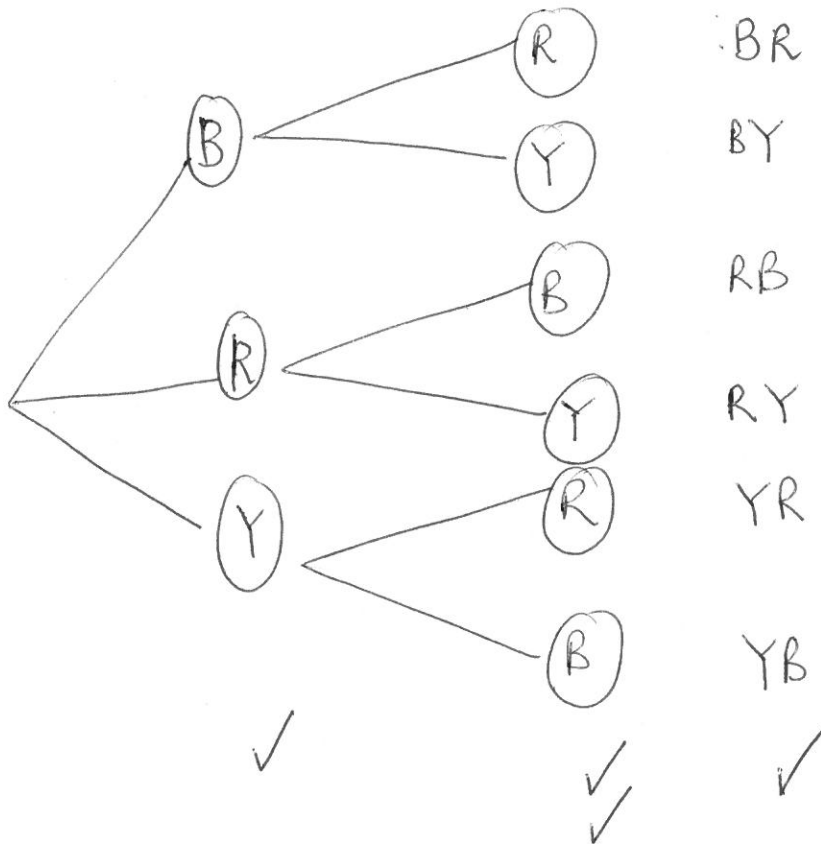


- e) a letter from the word SCIENCE

$\frac{4}{11}$



5. There are three marbles in a bag; 1 red, 1 blue and 1 yellow marble. We pull out one marble and place it on the bench. We then pull out a second marble and place that on the bench too. Draw a tree diagram to represent all of the possible combinations of marbles on the bench. List the outcomes on the side of the tree diagram. (4)



6. Complete the two way table below for students in Mr Torana's class about if they like apples and bananas. (6)

	Like bananas	Dislike bananas	Total
Like apples	7	16	23
Dislike apples	5	4	9
Total	12	20	32

Determine the probability that the students

- a) like apples and bananas.

$$\frac{7}{32} \checkmark$$

- b) like apples or bananas.

$$\frac{28}{32} \checkmark \checkmark$$

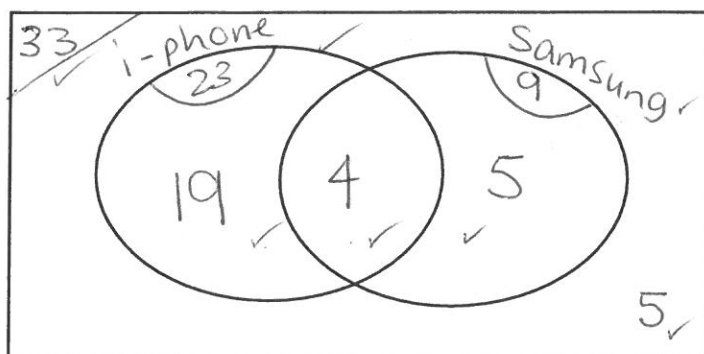
$$7 + 16 + 5 = 28 \text{ or } 32 - 4 = 28.$$

- c) don't like apples

$$\frac{9}{32} \checkmark \checkmark$$

7. In a class of 33 students, 23 have an iphone and 9 have a samsung. Four had both an iphone and a Samsung. Five students do not have phones.

a) Represent this using a Venn diagram. (4)



(8 half marks = 4)

b) What is the probability that a student has an iphone and a Samsung? (1)

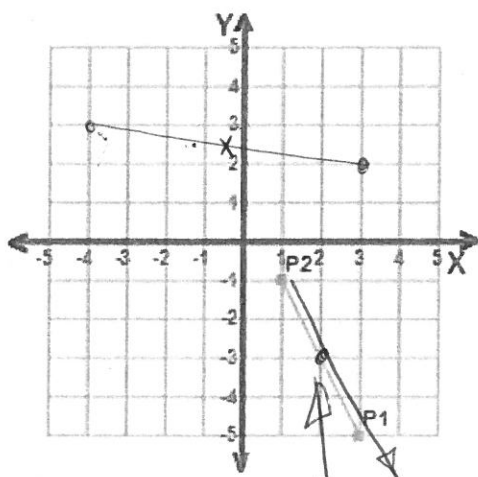
$$\frac{4}{33}$$

c) What is the probability that a student has an iphone? (1)

$$\frac{23}{33}$$

8. For the diagram below determine;

(6)



a) The distance between the two points.

$$C = \sqrt{4^2 + 2^2} \checkmark$$

$$C = \sqrt{20} \checkmark$$

$$= 4.47$$

b) The gradient of the line.

$$\frac{\text{rise}}{\text{run}} = \frac{-4}{2} = -2 \checkmark$$

c) The midpoint of P1 and P2.

$$\begin{matrix} x_1 & y_1 & x_2 & y_2 \\ (3, -5) & (1, -1) & & \end{matrix}$$

$$\frac{3-1}{2} = 1 \quad \frac{(-5)-(-1)}{2} = -2$$

9. Determine the midpoint of the points (3, 2) and (-4, 3).

(3)

$$x = \frac{3 + (-4)}{2} = -0.5 \checkmark$$

$$y = \frac{2 + 3}{2} = 2.5 \checkmark$$

$$\begin{matrix} x & y \\ 3 & 2 \\ -4 & 3 \end{matrix}$$

$$\frac{3 + (-4)}{2} = -0.5 \checkmark$$

$$\frac{2 + 3}{2} = 2.5 \checkmark$$

